

Integration Theory MATM39: Summary of Important Results

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Chapter 1

Sigma Algebras

Definition 1. Let X be an arbitrary set. A collection $\mathcal{A}$ of subsets of X is called a σ -algebra on X if:

- (a) $X \in \mathcal{A}$,
- (b) if $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$,
- (c) if $\{A_i\}_{i=1}^{\infty} \subset \mathcal{A}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$,
- (d) if $\{A_i\}_{i=1}^{\infty} \subset \mathcal{A}$, then $\bigcap_{i=1}^{\infty} A_i \in \mathcal{A}$.

Proof. Assume (a), (b), and (c). Let $\{A_i\}_{i=1}^{\infty} \subset \mathcal{A}$. Then $A_i^c \in \mathcal{A}$ for all i , and by (c), $\bigcup_{i=1}^{\infty} A_i^c \in \mathcal{A}$. By (b) again, $(\bigcup_{i=1}^{\infty} A_i^c)^c \in \mathcal{A}$. By De Morgan's law,

$$\bigcap_{i=1}^{\infty} A_i = \left(\bigcup_{i=1}^{\infty} A_i^c \right)^c \in \mathcal{A}$$

□

Proposition 1. Let X be a set. Then the intersection of an arbitrary nonempty collection of σ -algebras on X is a σ -algebra on X .

Proof. Let $\mathcal{C}$ be a nonempty collection of σ -algebras on X , and define

$$\mathcal{A} := \bigcap_{\mathcal{B} \in \mathcal{C}} \mathcal{B}.$$

Since every $\mathcal{B} \in \mathcal{C}$ contains X , we have $X \in \mathcal{A}$. If $A \in \mathcal{A}$, then $A \in \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$. Since each $\mathcal{B}$ is a σ -algebra, $A^c \in \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$, hence $A^c \in \mathcal{A}$. If $\{A_i\}_{i=1}^{\infty} \subset \mathcal{A}$, then $\{A_i\}_{i=1}^{\infty} \subset \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$. Hence

$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{B}$$

for every $\mathcal{B} \in \mathcal{C}$, and therefore

$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}.$$

Thus $\mathcal{A}$ is a σ -algebra on X .

□

Corollary 1. *Let X be a set, and let $\mathcal{F}$ be a family of subsets of X . Then there exists a smallest σ -algebra on X that contains $\mathcal{F}$.*

Proof. Let $\mathcal{C}$ be the collection of all σ -algebras on X that contain $\mathcal{F}$. This collection is nonempty, since $\mathcal{P}(X)$ is a σ -algebra on X and contains $\mathcal{F}$.

Define

$$\sigma(\mathcal{F}) := \bigcap_{\mathcal{A} \in \mathcal{C}} \mathcal{A}.$$

By the previous proposition, $\sigma(\mathcal{F})$ is a σ -algebra on X . It contains $\mathcal{F}$, since every $\mathcal{A} \in \mathcal{C}$ contains $\mathcal{F}$. Moreover, if $\mathcal{B}$ is any σ -algebra on X that contains $\mathcal{F}$, then $\mathcal{B} \in \mathcal{C}$, and therefore

$$\sigma(\mathcal{F}) \subset \mathcal{B}.$$

Thus $\sigma(\mathcal{F})$ is the smallest σ -algebra on X containing $\mathcal{F}$. □

Definition 2. *The Borel σ -algebra on $\mathbb{R}^d$ is the σ -algebra generated by the open subsets of $\mathbb{R}^d$. It is denoted by $\mathcal{B}(\mathbb{R}^d)$. Its elements are called Borel sets.*

Proposition 2. *The Borel σ -algebra $\mathcal{B}(\mathbb{R})$ is generated by each of the following collections:*

- (a) *the collection of all closed subsets of $\mathbb{R}$,*
- (b) *the collection of all intervals of the form $(-\infty, b]$,*
- (c) *the collection of all intervals of the form $(a, b]$.*

Proof. Let $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$ denote the σ -algebras generated by the collections in (a), (b), and (c), respectively.

Since $\mathcal{B}(\mathbb{R})$ contains all open sets and is closed under complements, it contains all closed sets. Hence

$$\mathcal{B}(\mathbb{R}) \supset \mathcal{B}_1.$$

Every set $(-\infty, b]$ is closed, so $\mathcal{B}_1 \supset \mathcal{B}_2$. Moreover,

$$(a, b] = (-\infty, b] \cap (-\infty, a]^c,$$

so $\mathcal{B}_2 \supset \mathcal{B}_3$.

Finally, let $G \subset \mathbb{R}$ be open. For each $x \in G$, since G is open, there exist rational numbers $a, b \in \mathbb{Q}$ such that

$$x \in (a, b] \subset G.$$

Hence

$$G = \bigcup_{\substack{a, b \in \mathbb{Q} \\ (a, b] \subset G}} (a, b].$$

This is a countable union, since $\mathbb{Q}^2$ is countable. Therefore $G \in \mathcal{B}_3$. Thus every open set belongs to $\mathcal{B}_3$, so

$$\mathcal{B}(\mathbb{R}) \subset \mathcal{B}_3.$$

Therefore all four σ -algebras are equal. □

Measures

Definition 3. Let X be a set and $\mathcal{A}$ a σ -algebra on X . A function $\mu : \mathcal{A} \rightarrow [0, \infty]$ is called a *measure* if

- (a) $\mu(\emptyset) = 0$,
- (b) for every $\{A_i\}_{i=1}^{\infty}$ of pairwise disjoint sets in $\mathcal{A}$, $\mu(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$.

Proposition 3. Let $(X, \mathcal{A}, \mu)$ be a measure space.

- (a) If $\{A_k\}$ is an increasing sequence in $\mathcal{A}$, then

$$\mu\left(\bigcup_{k=1}^{\infty} A_k\right) = \lim_{k \rightarrow \infty} \mu(A_k).$$

- (b) If $\{A_k\}$ is a decreasing sequence in $\mathcal{A}$ and $\mu(A_n) < \infty$ for some n , then

$$\mu\left(\bigcap_{k=1}^{\infty} A_k\right) = \lim_{k \rightarrow \infty} \mu(A_k).$$

Proof. (a) Define $B_1 = A_1$ and $B_k = A_k \setminus A_{k-1}$ for $k \geq 2$. Then $\{B_k\}$ are disjoint and $\bigcup_k A_k = \bigcup_k B_k$. Hence

$$\mu\left(\bigcup_k A_k\right) = \sum_k \mu(B_k) = \lim_{k \rightarrow \infty} \sum_{i=1}^k \mu(B_i) = \lim_{k \rightarrow \infty} \mu(A_k).$$

(b) Since $\mu(A_n) < \infty$ for some n , we may replace $\{A_k\}$ by the tail $\{A_{k+n-1}\}$ and assume $\mu(A_1) < \infty$. Let $C_k = A_1 \setminus A_k$. Then $\{C_k\}$ is increasing and

$$\bigcup_k C_k = A_1 \setminus \bigcap_k A_k.$$

By (a),

$$\mu(A_1 \setminus \bigcap_k A_k) = \lim_{k \rightarrow \infty} \mu(C_k) = \lim_{k \rightarrow \infty} (\mu(A_1) - \mu(A_k)),$$

which gives $\mu(\bigcap_k A_k) = \lim_k \mu(A_k)$. □

Outer Measures

Definition 4. Let X be a set. An *outer measure* on X is a function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ such that

- (a) $\mu^*(\emptyset) = 0$,
- (b) if $A \subset B$, then $\mu^*(A) \leq \mu^*(B)$,
- (c) for any sequence $\{A_n\}$ of subsets of X , $\mu^*(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mu^*(A_n)$.

Definition 5. The Lebesgue outer measure on $\mathbb{R}$ is the function $\lambda^* : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$ defined by

$$\lambda^*(A) = \inf \left\{ \sum_{i=1}^{\infty} (b_i - a_i) : A \subset \bigcup_{i=1}^{\infty} (a_i, b_i) \right\}.$$

Definition 6. The Lebesgue outer measure on $\mathbb{R}^d$ is defined by

$$\lambda^*(A) = \inf \left\{ \sum_{i=1}^{\infty} \text{vol}(R_i) : A \subset \bigcup_{i=1}^{\infty} R_i \right\},$$

where each R_i is a bounded open rectangle in $\mathbb{R}^d$ and

$$\text{vol}(I_1 \times \cdots \times I_d) = \prod_{j=1}^d |I_j|.$$

Definition 7. Let μ^* be an outer measure on X . A set $B \subset X$ is called μ^* -measurable if for all $A \subset X$,

$$\mu^*(A) = \mu^*(A \cap B) + \mu^*(A \cap B^c).$$

Remark 1. Outer measures are defined on all subsets of X , but are only subadditive in general. Carathéodory's criterion selects a class of sets on which μ^* becomes additive. This allows us to restrict μ^* to a σ -algebra and obtain a genuine measure (in particular, the Lebesgue measure).

Theorem 1. Let X be a set and μ^* an outer measure on X . Let

$$\mathcal{M}_{\mu^*} := \{B \subset X : B \text{ is } \mu^*\text{-measurable}\}.$$

Then:

- (a) $\mathcal{M}_{\mu^*}$ is a σ -algebra,
- (b) the restriction of μ^* to $\mathcal{M}_{\mu^*}$ is a measure.

Proof sketch. One shows that $\mathcal{M}_{\mu^*}$ is closed under complements directly from the definition. Closure under finite unions follows by applying the defining identity to B_1 and B_2 and decomposing sets appropriately.

Closure under countable unions is obtained by first proving the result for finite disjoint unions and then passing to the limit using subadditivity of μ^* .

Finally, countable additivity of μ^* on $\mathcal{M}_{\mu^*}$ follows by applying the defining property to disjoint measurable sets and using the previous decomposition. $\square$

Remark 2. Applying this theorem to the Lebesgue outer measure λ^* on $\mathbb{R}^d$ produces a σ -algebra $\mathcal{M}_{\lambda^*}$, called the Lebesgue σ -algebra. The restriction of λ^* to this σ -algebra is the Lebesgue measure. Thus, outer measure together with Carathéodory's criterion provides a systematic way to construct Lebesgue measure.

Example 1 (Cantor set). Let $K_0 = [0, 1]$. Construct K_n by removing the open middle third of each interval in K_{n-1} . The Cantor set is $K = \bigcap_{n=0}^{\infty} K_n$.

Proposition 4. The Cantor set K is compact, uncountable, and has Lebesgue measure zero.

Idea. Each K_n is a finite union of closed intervals, hence compact, and $K = \bigcap_n K_n$ is compact. One shows K has the cardinality of the continuum via ternary expansions.

Moreover, $\lambda(K_n) = (2/3)^n$, so

$$\lambda(K) \leq \lambda(K_n) = (2/3)^n \rightarrow 0,$$

hence $\lambda(K) = 0$. □

Theorem 2. *There exists a subset of $(0, 1)$ that is not Lebesgue measurable.*

Idea. Define $x \sim y$ if $x - y \in \mathbb{Q}$ and choose one representative from each equivalence class (using the axiom of choice) to form a set $E \subset (0, 1)$.

Consider the translates $E + r_n$ by rational numbers r_n . These sets are disjoint and their union covers $(0, 1)$ up to inclusion. Translation invariance and countable additivity would force a contradiction, so E cannot be measurable. □

Remark 3. *Not every subset of $\mathbb{R}$ can be assigned a consistent measure: the existence of non-measurable sets shows that countable additivity and translation invariance cannot hold on all subsets.*

Therefore, one must restrict attention to a σ -algebra (such as the Lebesgue σ -algebra) on which a well-defined measure exists.

Chapter 2

Chapter 3

Chapter 5